06- FINE STRUCTURE of TWO-ELECTRON ATOM He

PURPOSE of the EXPERIMENT

Determination of the wavelengths of the spectral lines of the two-electron helium (He) atom

GENERAL INFORMATION

In the ground state of the He atom, two s electrons fill the innermost shell with a principal quantum number $n = 1$. In the excited state, one electron remains in the half-filled first shell while the other is excited to the upper shell. Thus

1. Electron $l = 0$ in the case of $n = 1$ and

2. $l = 0...n\text{-}1$ if electron $n > 1$

has two electrons.

The term diagram (Grotrian diagram) for the helium atom is given in Figure 1. This diagram shows the allowed electronic transitions between the energy levels of atoms. They can be used for single-electron and multi-electron atoms and are constructed considering the selection rules. Helium in the singlet state is also called parahelium. Unlike alkali atoms, it has no fine structure. All its lines are single. The symbol 1^1S gives the lowest term. Here the first 1 is the head quantum number, the higher index 1 is the multiplicity (singlet) and the letter S stands for the total orbital angular momentum, which is zero. The larger terms are 2^1S, 2^1P, 3^1S, 3^1P, 3^1P, 3^1D, etc. Due to the lack of fine structure, they are oriented in such a way that the antiparallel and vectorial sum of the spins of the two electrons gives a total spin of $S = 0$. Unlike the single system, triplet helium has a fine structure and is called orthohelium. Its lowest level is 2^3S. Here 2 stands for the excited electron with $n = 2$, the top index 3 stands for the multiplicity term and the letter S stands for $L = 0$.

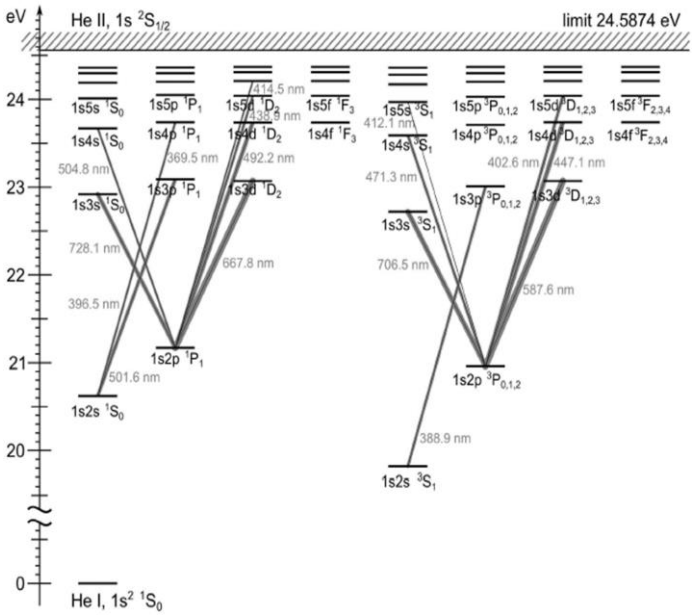


Figure 1: Term diagram of the He atom.

In this system, the spins of the two electrons are parallel to each other. Quantum number of total spin $s_1 + s_2 = S = 1$. Combinations between systems cannot be observed, i.e. there are no optical transitions between singlet and triplet systems. If one compares the mutual configurations in two non-combining systems, one finds that there are noticeable differences in energy, especially for low quantum numbers. The 2^1S_o state is 0.8 eV above the 2^3S_1 state and the 2^1P_1 state is around 0.25 eV above the 2^3P_2 state. This energy difference arises from the difference in electrostatic interactions between two electrons with parallel and antiparallel spin orientations.

The energy eigenvalues E in the atomic system can be found using the time-independent Schrödinger equation:

$$H\Psi(r_i) = E\Psi(r_i) \qquad (1)$$

Here Ψ is the wave function, r_i is the position of the i. electron, and H is the Hamiltonian energy operator and represents the total energy of the system. In many electron systems, this equation cannot be solved exactly because the interactions between electrons are not spherically symmetric. Some approximations are made to solve the equation.

When the spectra of many-electron atoms are analyzed, it is understood that the lines thought to be single lines are not single. As a result of the interaction between the spin and orbital moment of the electrons and the internal magnetic field, splits in the energy levels of the atom, called fine structure, occur. Electrons have an orbital angular momentum (L) together with an

emergent magnetic moment (μ_L) and a spin angular momentum (S) together with a magnetic moment (μ_S). The electron moves in the electric field of the nucleus and therefore has a magnetic field proportional to its angular momentum l. The orientational energy of the electron in the field is E_{nl}:

$$E_{nl} = -\boldsymbol{\mu}_S.\mathbf{B} = \zeta\, \boldsymbol{s}.\boldsymbol{l} \qquad (2)$$

ζ is the fine structure constant.

In many electron systems, the Hamiltonian is written as follows:

$$H = \sum_{i=1}^{N}\left(-\frac{\hbar^2}{2m}\nabla_i^2 - \frac{Ze^2}{(4\pi\varepsilon_0)r_i}\right) + \sum_{j>i=1}^{N}\frac{e^2}{(4\pi\varepsilon_0)r_{ij}} + \sum_{i=1}^{N}\zeta_i(r_i)\,\vec{s}_i\cdot\vec{\ell}_i \qquad (3)$$

The first term in this equation gives the kinetic energy of the electrons, the second term gives the Coulomb interaction between the nucleus and each electron, the third term gives the Coulomb interaction between electrons and the last term gives the spin-orbit interaction. If the interaction between the outer electrons is greater than the interaction of the spin and orbital moments for each electron, the LS Coupling (Russell-Saunders Coupling) is observed.

LS Coupling (Russell-Saunders Coupling)

The interaction between each angular momentum must be taken into account. The vector sum of spin and orbital angular momentum is given in the figure below

$$\boldsymbol{L} = \sum \boldsymbol{l}_i \text{ ve } \boldsymbol{S} = \sum \boldsymbol{s}_i \qquad (4)$$

is obtained. Values that the total angular momentum quantum number can take,

$$\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}, \qquad (5)$$

The combination of the electron configuration of an atom and the term symbol defines the energy eigenvalues of the atom. The quantum number L determines the term characteristics:

L = 0, 1, 2,.... corresponds to the terms

$S, P, D, \ldots$

When determining terms in LS coupling,

$$n^{2S+1}L_J \qquad (6)$$

Where n is the principal quantum number and $2S+1$ is the multiplicity term. The quantum number J takes the value $(2L+1)$ if $L < S$ and $(2S+1)$ if $L > S$. Each value of J corresponds to an energy level characterized by the possible orientations of the L and S vectors. Due to the mutual interactions of the total spin and orbital angular momenta of the outer electrons, the terms are decomposed into subterms, called multiplets, defined by the total J quantum number. Taking

the helium atom as an example, let us show that atomic terms can be derived from given electron configurations.

If both electrons are in the lowest shell, the electron configuration is 1s1s or $1s^2$ in the usual notation. In this case, they have the following quantum numbers:

$$n_1 = n_2 = 1, \quad l_1 = l_2 = 0, \quad s_1 = 1/2, \quad s_2 = 1/2$$

The resulting quantum numbers of the atom are then either ^1S_o for the singlet ground state

$$L = 0, S = 0 \qquad m_{s1} = - m_{s2}, \qquad J = 0$$

or $^3S_1$ for the triple ground state

$$L = 0, S = 1 \qquad m_{sl} = m_{s2}, \qquad J = 1$$

will be of the form. In reality, only a single ground state is observed. The triple ground state is forbidden by Pauli's principle since two electrons would not only have the same n and l quantum numbers but also the same m_s spin orientation. However, if one electron remains in the shell with $n = 1$ and the other is promoted to the state with $n = 2$ (1s2s electron configuration), we have the following quantum numbers:

$$n_1 = 1, n_2 = 2, \; l_1 = l_2 = 0, s_1 = 1/2, s_2 = 1/2$$

This gives either the singlet state $L = 0$, $S = 0$, $J = 0$, i.e. ^1S_o, or the triplet state $L = 0$, $S = 1$, $J = 1$, i.e. $^3S_1$. Both states are allowed and observed. In the same way, states and term symbols can be derived for all electron configurations.

Selection Rules

For a multi-electron system, the following selection rules apply:

$$\Delta L = 0, \pm 1$$

$$\Delta S = 0$$

$$\Delta J = 0, \pm 1 \qquad (J = 0 \rightarrow J = 0 \text{ Forbidden})$$

The spectral lines in Figure 1 are shown in Table 1 together with the term values of the lower and upper energy levels.

Table 1: Spectral lines of the He atom.

Color	λ/nm	Relative Transition	Relative Intensity
Red	706.5	$3\ ^3S \rightarrow 2\ ^1P$	5
Red	667.8	$3\ ^1D \rightarrow 2\ ^1P$	6
Red	656.0	He-II	4 – 6
Yellow	587.6	$3\ ^3D \rightarrow 2\ ^3P$	10
Green	504.8	$4\ ^1S \rightarrow 2\ ^1P$	2
Green	492.2	$4\ ^1D \rightarrow 2\ ^1P$	4
Blue	471.3	$4\ ^3S \rightarrow 2\ ^3P$	3
Blue	447.1	$4\ ^3D \rightarrow 2\ ^3P$	6
Blue	438.8	$5\ ^1D \rightarrow 2\ ^1P$	3
Purple	414.4	$6\ ^1D \rightarrow 2\ ^1P$	2
Purple	412.1	$5\ ^3S \rightarrow 2\ ^3P$	3
Purple	402.6	$5\ ^3D \rightarrow 2\ ^3P$	5
Purple	396.5	$4\ ^1P \rightarrow 2\ ^1S$	4
Purple	388.9	$3\ ^3P \rightarrow 2\ ^3S$	10

HOW TO CONDUCT THE EXPERIMENT



Figure 2: Experimental set for the calculation of the spectral lines of the helium atom.

As shown in Figure 2, the spectrometer consists of four main parts. These are the collimator, the goniometer, the area where the grating is placed, and the binoculars. The collimator has an adjustable slit spacing and parallelizes the beams.

If light of wavelength λ falls on a grating with a diffraction constant d, it is diffracted. Maximum intensities are found when the diffraction angle θ satisfies the following conditions for m = 0, 1, 2...:

$$m\,\lambda = d\,sin\theta$$

m is the order number. The value of the grating constant used in this experiment is d = 1684 nm.

In this experiment, the emission lines of a He lamp having two electrons will be analyzed. The excitation of He atoms with two electrons in the lamp media is made possible by electron collisions. The energy difference resulting from the excited state E_1 to the ground state E_o is emitted as a photon with frequency f:

$$hf = E_1 - E_0$$

Here h is Planck's constant ($h = 6.63.10^{-34}$ J.s).

1. Light the He lamp. The wavelengths of the visible colors of the He atom are given in Table 2.

Table 2: The spectral lines of the He atom that can be observed during the experiment whose wavelengths will be found, and their term values.

Color	λ/nm	Relative Transition	Relative Intensity
Red	667.8	$3\ ^1D \rightarrow 2\ ^1P$	6
Yellow	587.6	$3\ ^3D \rightarrow 2\ ^3P$	10
Green	504.8	$4\ ^1S \rightarrow 2\ ^1P$	2
Green	492.2	$4\ ^1D \rightarrow 2\ ^1P$	4
Blue	471.3	$4\ ^3S \rightarrow 2\ ^3P$	3
Blue	447.1	$4\ ^3D \rightarrow 2\ ^3P$	6

2. Place the grating on the table and adjust the table so that the light is perpendicular to it.

3. Set the binoculars to the zero bright line and read the angle on the goniometer.

4. Then turn the binoculars to the left and then to the right and read the values of θ_{right} and θ_{left} for the colors given in Table 2 at m = 1 and m = 2.

5. Subtract the zero-order angle you read in Step 3 from the diffraction angles θ_{left} and θ_{right} and enter them in Table 3.

6. For each spectral line in each color, calculate the average of the diffraction angles you found on the left and right sides.

7. Calculate the λ(nm) wavelengths of the colors you see using the formula $m\lambda = d\ (sin\theta_{ave})$.

8. Calculate the average of the wavelengths for each color found at $m = 1$ and $m = 2$ and record them in Table 3.

9. Draw the calibration curve of the spectrometer for the values of λ and $sin\theta_{ave}$ determined for $m = 1$.

Table 3: Results

m = 1

λ	θ_o	θ_{left}	$(\theta_{left}-\theta_o)$	θ_{right}	$(\theta_{right}-\theta_o)$	θ_{ave}	$sin\theta_{ave}$	λ(nm)
λ_{red}								
λ_{yellow}								
λ_{green}								
λ_{blue}								
λ_{purple}								

m = 2

λ	θ_o	θ_{left}	$(\theta_{left}-\theta_o)$	θ_{right}	$(\theta_{right}-\theta_o)$	θ_{ave}	$sin\theta_{ave}$	λ(nm)
λ_{red}								
λ_{yellow}								
λ_{green}								
λ_{blue}								
λ_{purple}								

	***m* = 1**	***m* = 2**	λ_{ave} (nm)
λ_{red}			
λ_{yellow}			
λ_{green}			
λ_{blue}			
λ_{purple}			

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